

Dietary habits and their relations to insulin resistance and postprandial lipemia in nonalcoholic steatohepatitis.



The relations of dietary habits to insulin sensitivity and postprandial triglyceride metabolism were evaluated in 25 patients with nonalcoholic steatohepatitis (NASH) and 25 age-, body mass index (BMI)-, and gender-matched healthy controls. After a 7-day alimentary record, they underwent a standard oral glucose tolerance test (OGTT), and the insulin sensitivity index (ISI) was calculated from the OGTT; an oral fat load test was also performed in 15 patients and 15 controls. The dietary intake of NASH patients was richer in saturated fat (13.7% $\pm$ 3.1% vs. 10.0% $\pm$ 2.1% total kcal, respectively, $P = .0001$) and in cholesterol (506 $\pm$ 108 vs. 405 $\pm$ 111 mg/d, respectively, $P = .002$) and was poorer in polyunsaturated fat (10.0% $\pm$ 3.5% vs. 14.5% $\pm$ 4.0% total fat, respectively, $P = .0001$), fiber (12.9 $\pm$ 4.1 vs. 23.2 $\pm$ 7.8 g/d, respectively, $P = .000$), and antioxidant vitamins C (84.3 $\pm$ 43.1 vs. 144.2 $\pm$ 63.1 mg/d, respectively, $P = .0001$) and E (5.4 $\pm$ 1.9 vs. 8.7 $\pm$ 2.9 mg/d, respectively, $P = .0001$). The ISI was significantly lower in NASH patients than in controls. Postprandial total and very low density lipoproteins triglyceride at +4 hours and +6 hours, triglyceride area under the curve, and incremental triglyceride area under the curve were higher in NASH compared with controls. Saturated fat intake correlated with ISI, with the different features of the metabolic syndrome, and with the postprandial rise of triglyceride. Postprandial apolipoprotein (Apo) B48 and ApoB100 responses in NASH were flat and strikingly dissociated from the triglyceride response, suggesting a defect in ApoB secretion. In conclusion, dietary habits may promote steatohepatitis directly by modulating hepatic triglyceride accumulation and antioxidant activity as well as indirectly by affecting insulin sensitivity and postprandial triglyceride metabolism. Our findings provide further rationale for more specific alimentary interventions, particularly in nonobese, nondiabetic normolipidemic NASH patients.